

Paper Code: ICT 202		Paper: Computer Graphics		L	T/P	C						
Paper ID:				3	-	3						
Prerequisite Paper: ICT209												
Marking Scheme :												
1. Teacher's Continuous Evaluation : 25 marks												
2. Term and Theory Examinations : 75 marks												
Instructions for paper setter												
1. There should be 9 questions in the term end examinations question paper.												
2. The first (1st) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type questions of total 15 marks.												
3. Apart from question one which is compulsory, rest of the paper shall consists of four units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain up to 5 sub-parts/sub-questions. Each unit shall have a marks weightage of 15.												
4. The questions are to be framed keeping in view the learning objectives of course/paper. The standard /level of the questions to be asked should be at the level of the prescribed text book.												
Course Objectives :												
1.	To introduce fundamentals of computer graphics, types of graphics and Raster graphics algorithms											
2.	To introduce geometric manipulation in 2D and 3D space, perspective projections, surface and solid modelling.											
3.	To understand Color models and various illumination models											
4.	To understand Rendering techniques and Advanced modelling techniques											
Course Outcomes (CO)												
CO 1	Ability to understand the usage of the computer graphics primitives and perform the operations on it like clipping etc.											
CO 2	Ability to perform any editing of operations on geometry of the objects through 2D and 3D transformations as per the requirements and should be able to model curves and surfaces using different techniques.											
CO 3	Ability to make the model appearance realistic in terms of desired color, material and final appearance calculations.											
CO 4	Ability to understand the concepts of different rendering techniques and advanced modelling techniques.											
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)												
	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	2	2	1	3
CO 2	3	2	2	2	2	-	-	-	2	2	1	3
CO 3	3	2	2	3	2	-	-	-	2	2	1	3
CO 4	3	2	2	3	2	-	-	-	2	2	1	3
UNIT - I												
Introduction to graphics and types of graphics, quality parameters of graphics display. Basic raster graphics algorithms for drawing 2 D primitives: DDA line, Bresenham's line, Bresenham's circle, midpoint circle, midpoint ellipse. Conic Sections, Clipping of line (Cohen Sutherland algorithm), clipping of polygon (Sutherland Hodgeman algorithm), polygon filling. Attributes of Output primitives, Antialiasing												
UNIT - II												
Geometric manipulation in 2D and 3D space, window to viewport transformations, homogeneous coordinates, projections: parallel and perspective projections.												
Generating curves like Hermite, Bezier and B-spline. Surface generation, wireframe, surface and solid modelling. 3-D polygon surfaces, polygon tables, polygon meshes.												
UNIT - III												
Visible surface determination techniques for visible surface determination: Z-buffer, A- buffer algorithm, scanline algorithm, area subdivision algorithm for implementation of hidden surface removal. Achromatic and hardware color models and software color models. Local and global illumination models calculations, Lambert, Gouraud & Phong shading techniques.												
UNIT - IV												
Rendering: introduction to ray casting, ray-tracing, recursive ray tracing, and shadows. Advanced procedural modelling: fractals, concept of fractals generation, concept of grammar-based modelling.												

Textbook(s):

1. D. D. Hearn, M.P. Baker, "Computer Graphics C version", Pearson Education India, 2nd Edition, 2002.
2. J.D. Foley et. al., "Computer Graphics Principles & Practice in C", Pearson Education India, 2nd Edition, 2006.

References:

1. R.H. Bartels, J.C. Beatty and B.A. Barsky, "An Introduction to Splines for use in Computer Graphics and Geometric Modeling", Morgan Kaufmann Publishers Inc., 1996.
2. W. M. Newman and R. F. Sproul, "Principles of Interactive Computer Graphics", McGraw-Hill Education, 2nd Edition, 2001.
3. Z. Xiang and R. Plastock, "Theory and Problems of Computer Graphics", Schaum's Series, McGraw Hill, 2nd Edition, 2017.
4. F.P. Preparata and M.I. Shamos, "Computational Geometry: An Introduction", Springer, Reprint of the original 1st ed. 1985 Edition, 2012.
5. D. Rogers and J. Adams, "Mathematical Elements for Computer Graphics", McGraw Hill Education, 2nd Edition, 2017.
6. David F. Rogers, "Procedural Elements for Computer Graphics", McGraw Hill Education, 2nd Edition, 2017.
7. Alan Watt and Mark Watt, "Advanced Animation and Rendering Techniques", Addison-Wesley, 2002.